

53. Show that the function $f(t) = 1/t^2$ does not possess a Laplace transform. [Hint: Write $\mathcal{L}\{1/t^2\}$ as two improper integrals:

$$\mathcal{L}\{1/t^2\} = \int_0^1 \frac{e^{-st}}{t^2} dt + \int_1^\infty \frac{e^{-st}}{t^2} dt = I_1 + I_2.$$

Show that I_1 diverges.]

54. If $\mathcal{L}\{f(t)\} = F(s)$ and $a > 0$ is a constant, show that

$$\mathcal{L}\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right).$$

This result is known as the **change of scale theorem**.

In Problems 55–58, use the given Laplace transform and the result in Problem 54 to find the indicated Laplace transform. Assume that a and k are positive constants.

55. $\mathcal{L}\{e^\eta\} = \frac{1}{s-1}$; $\mathcal{L}\{e^{a\eta}\}$

56. $\mathcal{L}\{\cos t\} = \frac{s}{s^2+1}$; $\mathcal{L}\{\cos kt\}$

57. $\mathcal{L}\{t - \sin t\} = \frac{1}{s^2(s^2+1)}$; $\mathcal{L}\{kt - \sin kt\}$

58. $\mathcal{L}\{\cos t \sinh t\} = \frac{s^2-2}{s^4+4}$; $\mathcal{L}\{\cos kt \sinh kt\}$

4.2 The Inverse Transform and Transforms of Derivatives

INTRODUCTION In this section we take a few small steps into an investigation of how the Laplace transform can be used to solve certain types of equations. After we discuss the concept of the *inverse* Laplace transform and examine the transforms of derivatives we then use the Laplace transform to solve some simple ordinary differential equations.

4.2.1 Inverse Transforms

The Inverse Problem If $F(s)$ represents the Laplace transform of a function $f(t)$, that is, $\mathcal{L}\{f(t)\} = F(s)$, we then say $f(t)$ is the **inverse Laplace transform** of $F(s)$ and write $f(t) = \mathcal{L}^{-1}\{F(s)\}$. For example, from Examples 1, 2, and 3 in Section 4.1 we have, respectively,

$$1 = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\}, \quad t = \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\}, \quad \text{and} \quad e^{-3t} = \mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\}.$$

The analogue of Theorem 4.1.1 for the inverse transform is presented next.

Theorem 4.2.1 Some Inverse Transforms

$$\begin{array}{ll} \text{(a)} & 1 = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} \\ \text{(b)} & t^n = \mathcal{L}^{-1}\left\{\frac{n!}{s^{n+1}}\right\}, n = 1, 2, 3, \dots \\ \text{(c)} & e^{at} = \mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} \\ \text{(d)} & \sin kt = \mathcal{L}^{-1}\left\{\frac{k}{s^2+k^2}\right\} \\ \text{(e)} & \cos kt = \mathcal{L}^{-1}\left\{\frac{s}{s^2+k^2}\right\} \\ \text{(f)} & \sinh kt = \mathcal{L}^{-1}\left\{\frac{k}{s^2-k^2}\right\} \\ \text{(g)} & \cosh kt = \mathcal{L}^{-1}\left\{\frac{s}{s^2-k^2}\right\} \end{array}$$

When evaluating inverse transforms, it often happens that a function of s under consideration does not match *exactly* the form of a Laplace transform $F(s)$ given in a table. It may be necessary to “fix up” the function of s by multiplying and dividing by an appropriate constant.

EXAMPLE 1 Applying Theorem 4.2.1

Evaluate (a) $\mathcal{L}^{-1}\left\{\frac{1}{s^5}\right\}$ (b) $\mathcal{L}^{-1}\left\{\frac{1}{s^2+7}\right\}$.

SOLUTION (a) To match the form given in part (b) of Theorem 4.2.1, we identify $n+1=5$ or $n=4$ and then multiply and divide by $4!$:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^5}\right\} = \frac{1}{4!} \mathcal{L}^{-1}\left\{\frac{4!}{s^5}\right\} = \frac{1}{24} t^4.$$

(b) To match the form given in part (d) of Theorem 4.2.1, we identify $k^2 = 7$ and so $k = \sqrt{7}$. We fix up the expression by multiplying and dividing by $\sqrt{7}$:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 7}\right\} = \frac{1}{\sqrt{7}} \mathcal{L}^{-1}\left\{\frac{\sqrt{7}}{s^2 + 7}\right\} = \frac{1}{\sqrt{7}} \sin \sqrt{7}t. \quad \equiv$$

|| $\mathcal{L}^{-1}$ Is a Linear Transform The inverse Laplace transform is also a linear transform; that is, for constants α and β ,

$$\mathcal{L}^{-1}\{\alpha F(s) + \beta G(s)\} = \alpha \mathcal{L}^{-1}\{F(s)\} + \beta \mathcal{L}^{-1}\{G(s)\}, \quad (1)$$

where F and G are the transforms of some functions f and g . Like (3) of Section 4.1, (1) extends to any finite linear combination of Laplace transforms.

EXAMPLE 2 Termwise Division and Linearity

Evaluate $\mathcal{L}^{-1}\left\{\frac{-2s + 6}{s^2 + 4}\right\}$.

SOLUTION We first rewrite the given function of s as two expressions by means of termwise division and then use (1):

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{-2s + 6}{s^2 + 4}\right\} &= \mathcal{L}^{-1}\left\{\frac{-2s}{s^2 + 4} + \frac{6}{s^2 + 4}\right\} = -2\mathcal{L}^{-1}\left\{\frac{s}{s^2 + 4}\right\} + \frac{6}{2}\mathcal{L}^{-1}\left\{\frac{2}{s^2 + 4}\right\} \quad \begin{array}{l} \text{termwise division} \\ \downarrow \end{array} \quad \begin{array}{l} \text{linearity and fixing up constants} \\ \downarrow \end{array} \\ &= -2\cos 2t + 3\sin 2t. \quad \leftarrow \text{parts (e) and (d) of Theorem 4.2.1 with } k = 2 \quad \equiv \end{aligned}$$

|| Partial Fractions Partial fractions play an important role in finding inverse Laplace transforms. As mentioned in Section 2.2, the decomposition of a rational expression into component fractions can be done quickly by means of a single command on most computer algebra systems. Indeed, some CASs have packages that implement Laplace transform and inverse Laplace transform commands. But for those of you without access to such software, we will review in this and subsequent sections some of the basic algebra in the important cases in which the denominator of a Laplace transform $F(s)$ contains distinct linear factors, repeated linear factors, and quadratic polynomials with no real factors. We shall examine each of these cases as this chapter develops.

EXAMPLE 3 Partial Fractions and Linearity

Partial fractions: distinct linear factors in denominator

Evaluate $\mathcal{L}^{-1}\left\{\frac{s^2 + 6s + 9}{(s - 1)(s - 2)(s + 4)}\right\}$.

SOLUTION There exist unique constants A , B , and C such that

$$\begin{aligned} \frac{s^2 + 6s + 9}{(s - 1)(s - 2)(s + 4)} &= \frac{A}{s - 1} + \frac{B}{s - 2} + \frac{C}{s + 4} \\ &= \frac{A(s - 2)(s + 4) + B(s - 1)(s + 4) + C(s - 1)(s - 2)}{(s - 1)(s - 2)(s + 4)}. \end{aligned}$$

Since the denominators are identical, the numerators are identical:

$$s^2 + 6s + 9 = A(s - 2)(s + 4) + B(s - 1)(s + 4) + C(s - 1)(s - 2). \quad (3)$$

By comparing coefficients of powers of s on both sides of the equality, we know that (3) is equivalent to a system of three equations in the three unknowns A , B , and C . However, recall that there is a shortcut for determining these unknowns. If we set $s = 1$, $s = 2$, and $s = -4$

in (3) we obtain, respectively,*

$$16 = A(-1)(5), \quad 25 = B(1)(6), \quad 1 = C(-5)(-6),$$

and so $A = -\frac{16}{5}$, $B = \frac{25}{6}$, $C = \frac{1}{30}$. Hence the partial fraction decomposition is

$$\frac{s^2 + 6s + 9}{(s-1)(s-2)(s+4)} = -\frac{16/5}{s-1} + \frac{25/6}{s-2} + \frac{1/30}{s+4}, \quad (4)$$

and thus, from the linearity of $\mathcal{L}^{-1}$ and part (c) of Theorem 4.2.1,

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{s^2 + 6s + 9}{(s-1)(s-2)(s+4)}\right\} &= -\frac{16}{5} \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} + \frac{25}{6} \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} + \frac{1}{30} \mathcal{L}^{-1}\left\{\frac{1}{s+4}\right\} \\ &= -\frac{16}{5} e^t + \frac{25}{6} e^{2t} + \frac{1}{30} e^{-4t}. \end{aligned} \quad (5) \equiv$$

4.2.2 Transforms of Derivatives

Transform of a Derivative As pointed out in the introduction to this chapter, our immediate goal is to use the Laplace transform to solve differential equations. To that end we need to evaluate quantities such as $\mathcal{L}\{dy/dt\}$ and $\mathcal{L}\{d^2y/dt^2\}$. For example, if f' is continuous for $t \geq 0$, then integration by parts gives

$$\begin{aligned} \mathcal{L}\{f'(t)\} &= \int_0^\infty e^{-st} f'(t) dt = e^{-st} f(t) \Big|_0^\infty + s \int_0^\infty e^{-st} f(t) dt \\ &= -f(0) + s \mathcal{L}\{f(t)\} \end{aligned}$$

or

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0). \quad (6)$$

Here we have assumed that $e^{-st} f(t) \rightarrow 0$ as $t \rightarrow \infty$. Similarly, with the aid of (6),

$$\begin{aligned} \mathcal{L}\{f''(t)\} &= \int_0^\infty e^{-st} f''(t) dt = e^{-st} f'(t) \Big|_0^\infty + s \int_0^\infty e^{-st} f'(t) dt \\ &= -f'(0) + s \mathcal{L}\{f'(t)\} \\ &= s[sF(s) - f(0)] - f'(0) \quad \leftarrow \text{from (6)} \end{aligned}$$

or

$$\mathcal{L}\{f''(t)\} = s^2 F(s) - sf(0) - f'(0). \quad (7)$$

In like manner it can be shown that

$$\mathcal{L}\{f'''(t)\} = s^3 F(s) - s^2 f(0) - sf'(0) - f''(0). \quad (8)$$

The recursive nature of the Laplace transform of the derivatives of a function f should be apparent from the results in (6), (7), and (8). The next theorem gives the Laplace transform of the n th derivative of f .

Theorem 4.2.2 Transform of a Derivative

If $f, f', \dots, f^{(n-1)}$ are continuous on $[0, \infty)$ and are of exponential order and if $f^{(n)}(t)$ is piecewise continuous on $[0, \infty)$, then

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0),$$

where $F(s) = \mathcal{L}\{f(t)\}$.

*The numbers 1, 2, and -4 are the zeros of the common denominator $(s-1)(s-2)(s+4)$.

Solving Linear ODEs It is apparent from the general result given in Theorem 4.2.2 that $\mathcal{L}\{d^n y/dt^n\}$ depends on $Y(s) = \mathcal{L}\{y(t)\}$ and the $n - 1$ derivatives of $y(t)$ evaluated at $t = 0$. This property makes the Laplace transform ideally suited for solving linear initial-value problems in which the differential equation has *constant coefficients*. Such a differential equation is simply a linear combination of terms $y, y', y'', \dots, y^{(n)}$:

$$a_n \frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \dots + a_0 y = g(t),$$

$$y(0) = y_0, y'(0) = y_1, \dots, y^{(n-1)}(0) = y_{n-1},$$

where the coefficients $a_i, i = 0, 1, \dots, n$ and $y_0, y_1, \dots, y_{n-1}$ are constants. By the linearity property, the Laplace transform of this linear combination is a linear combination of Laplace transforms:

$$a_n \mathcal{L}\left\{\frac{d^n y}{dt^n}\right\} + a_{n-1} \mathcal{L}\left\{\frac{d^{n-1} y}{dt^{n-1}}\right\} + \dots + a_0 \mathcal{L}\{y\} = \mathcal{L}\{g(t)\}. \quad (9)$$

From Theorem 4.2.2, (9) becomes

$$a_n[s^n Y(s) - s^{n-1}y(0) - \dots - y^{(n-1)}(0)]$$

$$+ a_{n-1}[s^{n-1}Y(s) - s^{n-2}y(0) - \dots - y^{(n-2)}(0)] + \dots + a_0 Y(s) = G(s), \quad (10)$$

where $\mathcal{L}\{y(t)\} = Y(s)$ and $\mathcal{L}\{g(t)\} = G(s)$. In other words:

The Laplace transform of a linear differential equation with constant coefficients becomes an algebraic equation in $Y(s)$.

If we solve the general transformed equation (10) for the symbol $Y(s)$, we first obtain $P(s)Y(s) = Q(s) + G(s)$, and then write

$$Y(s) = \frac{Q(s)}{P(s)} + \frac{G(s)}{P(s)}, \quad (11)$$

where $P(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_0$, $Q(s)$ is a polynomial in s of degree less than or equal to $n - 1$ consisting of the various products of the coefficients $a_i, i = 1, \dots, n$, and the prescribed initial conditions $y_0, y_1, \dots, y_{n-1}$, and $G(s)$ is the Laplace transform of $g(t)$.^{*} Typically we put the two terms in (11) over the least common denominator and then decompose the expression into two or more partial fractions. Finally, the solution $y(t)$ of the original initial-value problem is $y(t) = \mathcal{L}^{-1}\{Y(s)\}$, where the inverse transform is done term by term.

The procedure is summarized in **FIGURE 4.2.1**.

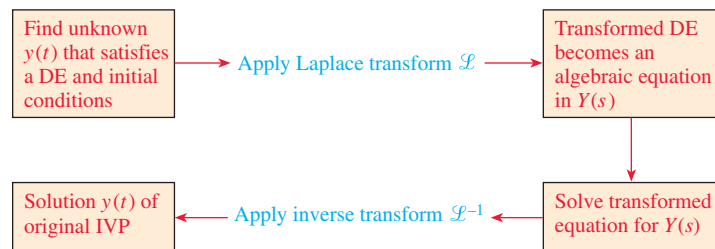


FIGURE 4.2.1 Steps in solving an IVP by the Laplace transform

The next example illustrates the foregoing method of solving DEs.

EXAMPLE 4 Solving a First-Order IVP

Use the Laplace transform to solve the initial-value problem

$$\frac{dy}{dt} + 3y = 13 \sin 2t, \quad y(0) = 6.$$

^{*}The polynomial $P(s)$ is the same as the n th degree auxiliary polynomial in (13) in Section 3.3, with the usual symbol m replaced by s .

SOLUTION We first take the transform of each member of the differential equation:

$$\mathcal{L}\left\{\frac{dy}{dt}\right\} + 3\mathcal{L}\{y\} = 13\mathcal{L}\{\sin 2t\}. \quad (12)$$

But from (6), $\mathcal{L}\{dy/dt\} = sY(s) - y(0) = sY(s) - 6$, and from part (d) of Theorem 4.1.1, $\mathcal{L}\{\sin 2t\} = 2/(s^2 + 4)$, and so (12) is the same as

$$sY(s) - 6 + 3Y(s) = \frac{26}{s^2 + 4} \quad \text{or} \quad (s + 3)Y(s) = 6 + \frac{26}{s^2 + 4}.$$

Solving the last equation for $Y(s)$, we get

$$Y(s) = \frac{6}{s + 3} + \frac{26}{(s + 3)(s^2 + 4)} = \frac{6s^2 + 50}{(s + 3)(s^2 + 4)}. \quad (13)$$

Partial fractions:
quadratic polynomial
with no real factors

Since the quadratic polynomial $s^2 + 4$ does not factor using real numbers, its assumed numerator in the partial fraction decomposition is a linear polynomial in s :

$$\frac{6s^2 + 50}{(s + 3)(s^2 + 4)} = \frac{A}{s + 3} + \frac{Bs + C}{s^2 + 4}.$$

Putting the right side of the equality over a common denominator and equating numerators gives $6s^2 + 50 = A(s^2 + 4) + (Bs + C)(s + 3)$. Setting $s = -3$ then yields immediately $A = 8$. Since the denominator has no more real zeros, we equate the coefficients of s^2 and s : $6 = A + B$ and $0 = 3B + C$. Using the value of A in the first equation gives $B = -2$, and then using this last value in the second equation gives $C = 6$. Thus

$$Y(s) = \frac{6s^2 + 50}{(s + 3)(s^2 + 4)} = \frac{8}{s + 3} + \frac{-2s + 6}{s^2 + 4}.$$

We are not quite finished because the last rational expression still has to be written as two fractions. But this was done by termwise division in Example 2. From (2) of that example,

$$y(t) = 8\mathcal{L}^{-1}\left\{\frac{1}{s + 3}\right\} - 2\mathcal{L}^{-1}\left\{\frac{s}{s^2 + 4}\right\} + 3\mathcal{L}^{-1}\left\{\frac{2}{s^2 + 4}\right\}.$$

It follows from parts (c), (d), and (e) of Theorem 4.2.1 that the solution of the initial-value problem is $y(t) = 8e^{-3t} - 2\cos 2t + 3\sin 2t$. $\equiv$

EXAMPLE 5 Solving a Second-Order IVP

Solve $y'' - 3y' + 2y = e^{-4t}$, $y(0) = 1$, $y'(0) = 5$.

SOLUTION Proceeding as in Example 4, we transform the DE by taking the sum of the transforms of each term, use (6) and (7), use the given initial conditions, use part (c) of Theorem 4.1.1, and then solve for $Y(s)$:

$$\begin{aligned} \mathcal{L}\left\{\frac{d^2y}{dt^2}\right\} - 3\mathcal{L}\left\{\frac{dy}{dt}\right\} + 2\mathcal{L}\{y\} &= \mathcal{L}\{e^{-4t}\} \\ s^2Y(s) - sy(0) - y'(0) - 3[sY(s) - y(0)] + 2Y(s) &= \frac{1}{s + 4} \\ (s^2 - 3s + 2)Y(s) &= s + 2 + \frac{1}{s + 4} \\ Y(s) &= \frac{s + 2}{s^2 - 3s + 2} + \frac{1}{(s^2 - 3s + 2)(s + 4)} = \frac{s^2 + 6s + 9}{(s - 1)(s - 2)(s + 4)} \end{aligned} \quad (14)$$

and so $y(t) = \mathcal{L}^{-1}\{Y(s)\}$. The details of the decomposition of $Y(s)$ in (14) into partial fractions have already been carried out in Example 3. In view of (4) and (5) the solution of the initial-value problem is $y(t) = -\frac{16}{5}e^t + \frac{25}{6}e^{2t} + \frac{1}{30}e^{-4t}$. $\equiv$

Examples 4 and 5 illustrate the basic procedure for using the Laplace transform to solve a linear initial-value problem, but these examples may appear to demonstrate a method that is not much better than the approach to such problems outlined in Sections 2.3 and 3.3–3.6. Don't draw any negative conclusions from the two examples. Yes, there is a lot of algebra inherent in the use of the Laplace transform, *but* observe that we do not have to use variation of parameters or worry about the cases and algebra in the method of undetermined coefficients. Moreover, since the method incorporates the prescribed initial conditions directly into the solution, there is no need for the separate operations of applying the initial conditions to the general solution $y = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n + y_p$ of the DE to find specific constants in a particular solution of the IVP.

The Laplace transform has many operational properties. We will examine some of these properties and illustrate how they enable us to solve problems of greater complexity in the sections that follow.

We conclude this section with a little bit of additional theory related to the types of functions of s that we will generally be working with. The next theorem indicates that not every arbitrary function of s is a Laplace transform of a piecewise-continuous function of exponential order.

Theorem 4.2.3 Behavior of $F(s)$ as $s \rightarrow \infty$

If f is piecewise continuous on $[0, \infty)$ and of exponential order, then $\lim_{s \rightarrow \infty} \mathcal{L}\{f(t)\} = 0$.

PROOF: Since $f(t)$ is piecewise continuous on the closed interval $[0, T]$, it is necessarily bounded on the interval. That is, $|f(t)| \leq M_1 = M_1 e^{0t}$. Also, because f is assumed to be of exponential order, there exist constants $\gamma, M_2 > 0$, and $T > 0$, such that $|f(t)| \leq M_2 e^{\gamma t}$ for $t > T$. If M denotes the maximum of $\{M_1, M_2\}$ and c denotes the maximum of $\{0, \gamma\}$, then

$$|\mathcal{L}\{f(t)\}| \leq \int_0^\infty e^{-st} |f(t)| dt \leq M \int_0^\infty e^{-st} \cdot e^{ct} dt = -M \frac{e^{-(s-c)t}}{s-c} \Big|_0^\infty = \frac{M}{s-c}$$

for $s > c$. As $s \rightarrow \infty$, we have $|\mathcal{L}\{f(t)\}| \rightarrow 0$, and so $\mathcal{L}\{f(t)\} \rightarrow 0$. ≡

As a consequence of Theorem 4.2.3 we can say that functions of s such as $F_1(s) = 1$ and $F_2(s) = s/(s+1)$ are not the Laplace transforms of piecewise-continuous functions of exponential order since $F_1(s) \not\rightarrow 0$ and $F_2(s) \not\rightarrow 0$ as $s \rightarrow \infty$. But you should not conclude from this that $F_1(s)$ and $F_2(s)$ are not Laplace transforms. There are other kinds of functions.

REMARKS

(i) The inverse Laplace transform of a function $F(s)$ may not be unique; in other words, it is possible that $\mathcal{L}\{f_1(t)\} = \mathcal{L}\{f_2(t)\}$ and yet $f_1 \neq f_2$. For our purposes this is not anything to be concerned about. If f_1 and f_2 are piecewise continuous on $[0, \infty)$ and of exponential order, then f_1 and f_2 are *essentially* the same. See Problem 50 in Exercises 4.2. However, if f_1 and f_2 are continuous on $[0, \infty)$ and $\mathcal{L}\{f_1(t)\} = \mathcal{L}\{f_2(t)\}$, then $f_1 = f_2$ on the interval.

(ii) This remark is for those of you who will be required to do partial fraction decompositions by hand. There is another way of determining the coefficients in a partial fraction decomposition in the special case when $\mathcal{L}\{f(t)\} = F(s)$ is a rational function of s and the denominator of F is a product of *distinct* linear factors. Let us illustrate by reexamining Example 3. Suppose we multiply both sides of the assumed decomposition

$$\frac{s^2 + 6s + 9}{(s-1)(s-2)(s+4)} = \frac{A}{s-1} + \frac{B}{s-2} + \frac{C}{s+4} \quad (15)$$

by, say, $s - 1$, simplify, and then set $s = 1$. Since the coefficients of B and C on the right side of the equality are zero, we get

$$\left. \frac{s^2 + 6s + 9}{(s - 2)(s + 4)} \right|_{s=1} = A \quad \text{or} \quad A = -\frac{16}{5}.$$

Written another way,

$$\frac{s^2 + 6s + 9}{\boxed{(s - 1)}(s - 2)(s + 4)} \bigg|_{s=1} = -\frac{16}{5} = A,$$

where we have colored or *covered up* the factor that canceled when the left side was multiplied by $s - 1$. Now to obtain B and C we simply evaluate the left-hand side of (15) while covering up, in turn, $s - 2$ and $s + 4$:

$$\left. \frac{s^2 + 6s + 9}{(s - 1)\boxed{(s - 2)}(s + 4)} \right|_{s=2} = \frac{25}{6} = B \quad \text{and} \quad \left. \frac{s^2 + 6s + 9}{(s - 1)(s - 2)\boxed{(s + 4)}} \right|_{s=-4} = \frac{1}{30} = C.$$

The desired decomposition (15) is given in (4). This special technique for determining coefficients is naturally known as the **cover-up method**.

(iii) In this remark we continue our introduction to the terminology of dynamical systems. Because of (9) and (10) the Laplace transform is well adapted to *linear* dynamical systems. In (11) the polynomial $P(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0$ is the total coefficient of $Y(s)$ in (10) and is simply the left-hand side of the DE with the derivatives $d^k y/dt^k$ replaced by powers s^k , $k = 0, 1, \dots, n$. It is usual practice to call the reciprocal of $P(s)$, namely, $W(s) = 1/P(s)$, the **transfer function** of the system and write (11) as

$$Y(s) = W(s)Q(s) + W(s)G(s). \quad (16)$$

In this manner we have separated, in an additive sense, the effects on the response that are due to the initial conditions (that is, $W(s)Q(s)$) and to the input function g (that is, $W(s)G(s)$). See (13) and (14). Hence the response $y(t)$ of the system is a superposition of two responses

$$y(t) = \mathcal{L}^{-1}\{W(s)Q(s)\} + \mathcal{L}^{-1}\{W(s)G(s)\} = y_0(t) + y_1(t).$$

If the input is $g(t) = 0$, then the solution of the problem is $y_0(t) = \mathcal{L}^{-1}\{W(s)Q(s)\}$. This solution is called the **zero-input response** of the system. On the other hand, the function $y_1(t) = \mathcal{L}^{-1}\{W(s)G(s)\}$ is the output due to the input $g(t)$. Now if the initial state of the system is the zero state (all the initial conditions are zero), then $Q(s) = 0$, and so the only solution of the initial-value problem is $y_1(t)$. The latter solution is called the **zero-state response** of the system. Both $y_0(t)$ and $y_1(t)$ are particular solutions: $y_0(t)$ is a solution of the IVP consisting of the associated homogeneous equation with the given initial conditions, and $y_1(t)$ is a solution of the IVP consisting of the nonhomogeneous equation with zero initial conditions. In Example 5, we see from (14) that the transfer function is $W(s) = 1/(s^2 - 3s + 2)$, the zero-input response is

$$y_0(t) = \mathcal{L}^{-1}\left\{\frac{s + 2}{(s - 1)(s - 2)}\right\} = -3e^t + 4e^{2t},$$

and the zero-state response is

$$y_1(t) = \mathcal{L}^{-1}\left\{\frac{1}{(s - 1)(s - 2)(s + 4)}\right\} = -\frac{1}{5}e^t + \frac{1}{6}e^{2t} + \frac{1}{30}e^{-4t}.$$

Verify that the sum of $y_0(t)$ and $y_1(t)$ is the solution $y(t)$ in that example and that $y_0(0) = 1$, $y_0'(0) = 5$, whereas $y_1(0) = 0$, $y_1'(0) = 0$.

4.2.1 Inverse Transforms

In Problems 1–30, use Theorem 4.2.1 to find the given inverse transform.

1. $\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\}$
2. $\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\}$
3. $\mathcal{L}^{-1}\left\{\frac{1}{s^2} - \frac{48}{s^5}\right\}$
4. $\mathcal{L}^{-1}\left\{\left(\frac{2}{s} - \frac{1}{s^3}\right)^2\right\}$
5. $\mathcal{L}^{-1}\left\{\frac{(s+1)^3}{s^4}\right\}$
6. $\mathcal{L}^{-1}\left\{\frac{(s+2)^2}{s^3}\right\}$
7. $\mathcal{L}^{-1}\left\{\frac{1}{s^2} - \frac{1}{s} + \frac{1}{s-2}\right\}$
8. $\mathcal{L}^{-1}\left\{\frac{4}{s} + \frac{6}{s^5} - \frac{1}{s+8}\right\}$
9. $\mathcal{L}^{-1}\left\{\frac{1}{4s+1}\right\}$
10. $\mathcal{L}^{-1}\left\{\frac{1}{5s-2}\right\}$
11. $\mathcal{L}^{-1}\left\{\frac{5}{s^2+49}\right\}$
12. $\mathcal{L}^{-1}\left\{\frac{10s}{s^2+16}\right\}$
13. $\mathcal{L}^{-1}\left\{\frac{4s}{4s^2+1}\right\}$
14. $\mathcal{L}^{-1}\left\{\frac{1}{4s^2+1}\right\}$
15. $\mathcal{L}^{-1}\left\{\frac{2s-6}{s^2+9}\right\}$
16. $\mathcal{L}^{-1}\left\{\frac{s+1}{s^2+2}\right\}$
17. $\mathcal{L}^{-1}\left\{\frac{1}{s^2+3s}\right\}$
18. $\mathcal{L}^{-1}\left\{\frac{s+1}{s^2-4s}\right\}$
19. $\mathcal{L}^{-1}\left\{\frac{s}{s^2+2s-3}\right\}$
20. $\mathcal{L}^{-1}\left\{\frac{1}{s^2+s-20}\right\}$
21. $\mathcal{L}^{-1}\left\{\frac{0.9s}{(s-0.1)(s+0.2)}\right\}$
22. $\mathcal{L}^{-1}\left\{\frac{s-3}{(s-\sqrt{3})(s+\sqrt{3})}\right\}$
23. $\mathcal{L}^{-1}\left\{\frac{s}{(s-2)(s-3)(s-6)}\right\}$
24. $\mathcal{L}^{-1}\left\{\frac{s^2+1}{s(s-1)(s+1)(s-2)}\right\}$
25. $\mathcal{L}^{-1}\left\{\frac{1}{s^3+5s}\right\}$
26. $\mathcal{L}^{-1}\left\{\frac{s}{(s+2)(s^2+4)}\right\}$
27. $\mathcal{L}^{-1}\left\{\frac{2s-4}{(s^2+s)(s^2+1)}\right\}$
28. $\mathcal{L}^{-1}\left\{\frac{1}{s^4-9}\right\}$
29. $\mathcal{L}^{-1}\left\{\frac{1}{(s^2+1)(s^2+4)}\right\}$
30. $\mathcal{L}^{-1}\left\{\frac{6s+3}{s^4+5s^2+4}\right\}$

In Problems 31 and 32, find the given inverse Laplace transform by finding the Laplace transform of the indicated function f .

31. $\mathcal{L}^{-1}\left\{\frac{1}{(s^2+a^2)(s^2+b^2)}\right\}$; $f(t) = a \sin bt - b \sin at$
32. $\mathcal{L}^{-1}\left\{\frac{s}{(s^2+a^2)(s^2+b^2)}\right\}$; $f(t) = \cos bt - \cos at$

4.2.2 Transforms of Derivatives

In Problems 33–44, use the Laplace transform to solve the given initial-value problem.

33. $\frac{dy}{dt} - y = 1, \quad y(0) = 0$
34. $2\frac{dy}{dt} + y = 0, \quad y(0) = -3$
35. $y' + 6y = e^{4t}, \quad y(0) = 2$
36. $y' - y = 2 \cos 5t, \quad y(0) = 0$
37. $y'' + 5y' + 4y = 0, \quad y(0) = 1, \quad y'(0) = 0$
38. $y'' - 4y' = 6e^{3t} - 3e^{-t}, \quad y(0) = 1, \quad y'(0) = -1$
39. $y'' + y = \sqrt{2} \sin \sqrt{2}t, \quad y(0) = 10, \quad y'(0) = 0$
40. $y'' + 9y = e^t, \quad y(0) = 0, \quad y'(0) = 0$
41. $2y''' + 3y'' - 3y' - 2y = e^{-t}, \quad y(0) = 0, \quad y'(0) = 0, \quad y''(0) = 1$
42. $y''' + 2y'' - y' - 2y = \sin 3t, \quad y(0) = 0, \quad y'(0) = 0, \quad y''(0) = 1$
43. $y^{(4)} - y = 0, \quad y(0) = 0, \quad y'(0) = 0, \quad y''(0) = 0, \quad y'''(0) = 2$
44. $y^{(4)} - 9y = 0, \quad y(0) = 1, \quad y'(0) = 0, \quad y''(0) = 3, \quad y'''(0) = 0$

In Problems 45 and 46, use one of the inverse Laplace transforms found in Problems 31 and 32 to solve the given initial-value problem.

45. $y'' + 4y = 10 \cos 5t, \quad y(0) = 0, \quad y'(0) = 0$
46. $y'' + 2y = \sin t, \quad y(0) = 0, \quad y'(0) = 0$

The inverse forms of the results in Problem 49 in Exercises 4.1 are

$$\mathcal{L}^{-1}\left\{\frac{s-a}{(s-a)^2+b^2}\right\} = e^{at} \cos bt$$

and
$$\mathcal{L}^{-1}\left\{\frac{b}{(s-a)^2+b^2}\right\} = e^{at} \sin bt.$$

In Problems 47 and 48, use the Laplace transform and these inverses to solve the given initial-value problem.

47. $y' + y = e^{-3t} \cos 2t, \quad y(0) = 0$
48. $y'' - 2y' + 5y = 0, \quad y(0) = 1, \quad y'(0) = 3$

Discussion Problems

49. (a) With a slight change in notation the transform in (6) is the same as

$$\mathcal{L}\{f'(t)\} = s \mathcal{L}\{f(t)\} - f(0).$$

With $f(t) = te^{at}$, discuss how this result in conjunction with part (c) of Theorem 4.1.1 can be used to evaluate $\mathcal{L}\{te^{at}\}$.

- (b) Proceed as in part (a), but this time discuss how to use (7) with $f(t) = t \sin kt$ in conjunction with parts (d) and (e) of Theorem 4.1.1 to evaluate $\mathcal{L}\{t \sin kt\}$.